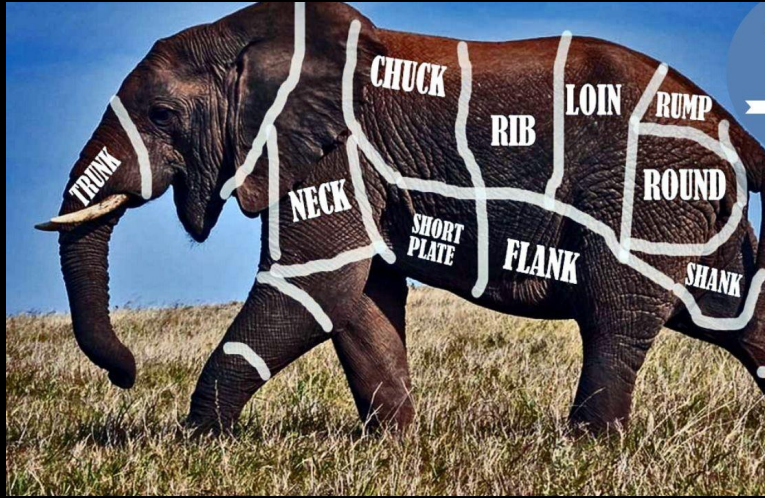




Part II:

How to eat an elephant

Three aims:

1. Start to build key **skills** to use AI effectively and efficiently
2. Understand the **risks and quirks**
3. Outline and plan **steps** to address risks around bias and accuracy

Recap: biggest concerns

1. Jobs and skills
2. Bias and accuracy
3. Environmental
4. Data protection and privacy
5. Power and accountability
6. Creativity

Questions: jobs

1. How do we upskill?
2. How can AI help improve efficiency?
3. How do we build on what we do?
4. How do we keep up with industry practices?
5. Which bits of the job would AI be useful for?
6. How do we avoid deskilling where AI is regularly used?
7. Job satisfaction: will people love ability to do research?

1. What can we do?

“Which bits of the job?”

Pre-production

Idea development
Lead identification
Stakeholder mapping
System mapping
Background research
Planning (general,
interviews, shoots,
outlines, simulations, etc.)

Production

Transcription
Translation
Generation (text,
images, audio, video)
Drafting inc. emails
Coding
Classification
Analysis

Post-production

Reviewing: diversity, style,
accessibility
Editing: spelling, grammar,
jargon, structure, brevity,
optimisation
Reversioning/personalisation
Distribution
Debugging, testing

“Some traditional content creation roles may diminish, [and] new opportunities will likely appear. These could include **content creators skilled in using LLMs**, who might earn higher wages, as well as **supervisory roles like editors, curators, and LLM system managers**. There will also be a need for **technicians to maintain and improve LLM systems**.”

“How do we upskill?”

- **Prompt design** skills
- **Risk assessment** and management skills
- **Prompt templating** skills (embedding knowledge)
- **Verification** and testing skills
- **Content review** and adaptation skills

Getting hands-on

- Ask your AI tool to 'Give me an idea for a story'
- What assumptions does it make about what you mean?
- Why?

CAREful prompts

- **Context:** describe the situation (role, employer, events)
- **Ask:** request specific action
- **Rules:** provide constraints (audience, genre, length)
- **Examples:** demonstrate what you want
- + Iterate, refine, and combine outputs

Add CARE

- What context can you add?
- What rules?
- Can you be clearer about what you are asking it to do?
- Can you give it an example?
- Can you iterate (give feedback to inform results)?

Prompt design techniques

- **Role prompting**: “You are a social media editor”
- **Recursive prompting**: “More like number 2 please”
- **N-shot prompting**: “Here are two examples”
- **RAG (Retrieval Augmented Generation)**: “Factor in this information...”
- **CoT (Chain of Thought)**: “Begin by doing X, then Y”
- **Meta-prompting**: “Suggest a prompt”
- **Negative prompting**: “Don’t suggest further actions”

A scenario

Design prompts for one of the following scenarios:

- **Research** for a briefing on an MP
- **Summarise** a public consultation for an internal report
- **Plan** and structure an internal presentation
- **Convert** a report into a series of social media posts
- **Analyse** data on user behaviour

Things to consider:

- What role might you give?
- What feedback?
- What examples might help guide the results?
- What information is needed to inform it?
- What steps should it take?
- What do you *not* want it to do?
- What prompts does AI itself suggest?

Chain-of-Thought prompting example

Prepare a background briefing on Peter Kyle, for a human rights charity assessing how government leadership may influence technology policy and its human rights implications. Think step by step.

Begin by outlining Peter Kyle's current ministerial role and responsibilities.

Then, examine his political background, including party affiliation, previous roles, and relevant committee work.

Identify key policy positions or legislative actions he has taken, particularly those connected to digital rights, AI regulation, privacy, or freedom of expression.

Consider any public statements, voting records or affiliations with organisations that may indicate his stance on human rights issues in a technological context.

Finally, summarise how his profile may influence the government's approach to science and technology as it relates to human rights."

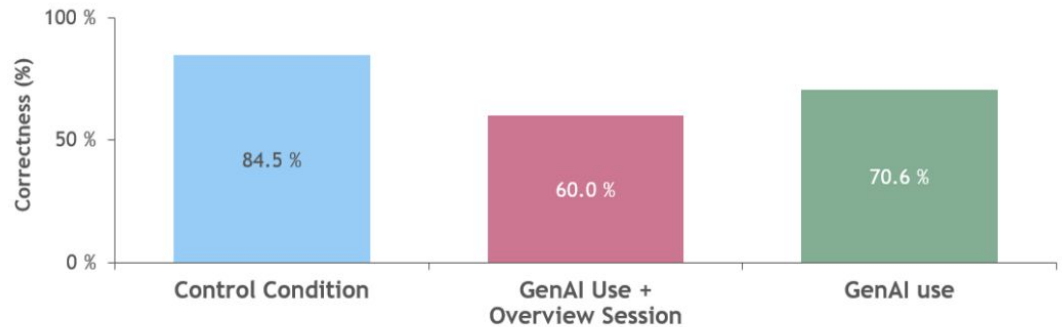
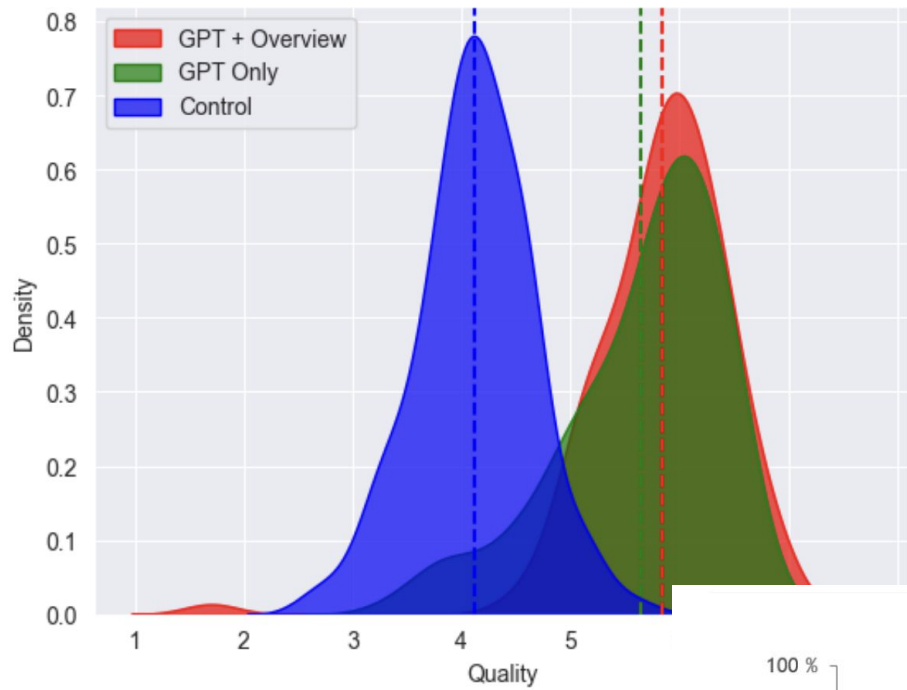
2. Risks and quirks

AI involves a lot of risks...

- **Accuracy**: AI hallucinates, including citations, *confidently*
- **Biases**: demographic, algorithmic, temporal, data-based
- **Environmental**: energy-intensive
- **Employment**: destroying/creating/displacing jobs
- **Privacy/security/copyright**: some models [store the data](#)
- **Explainability**: do you know how it arrived at the result?

...and it has important quirks

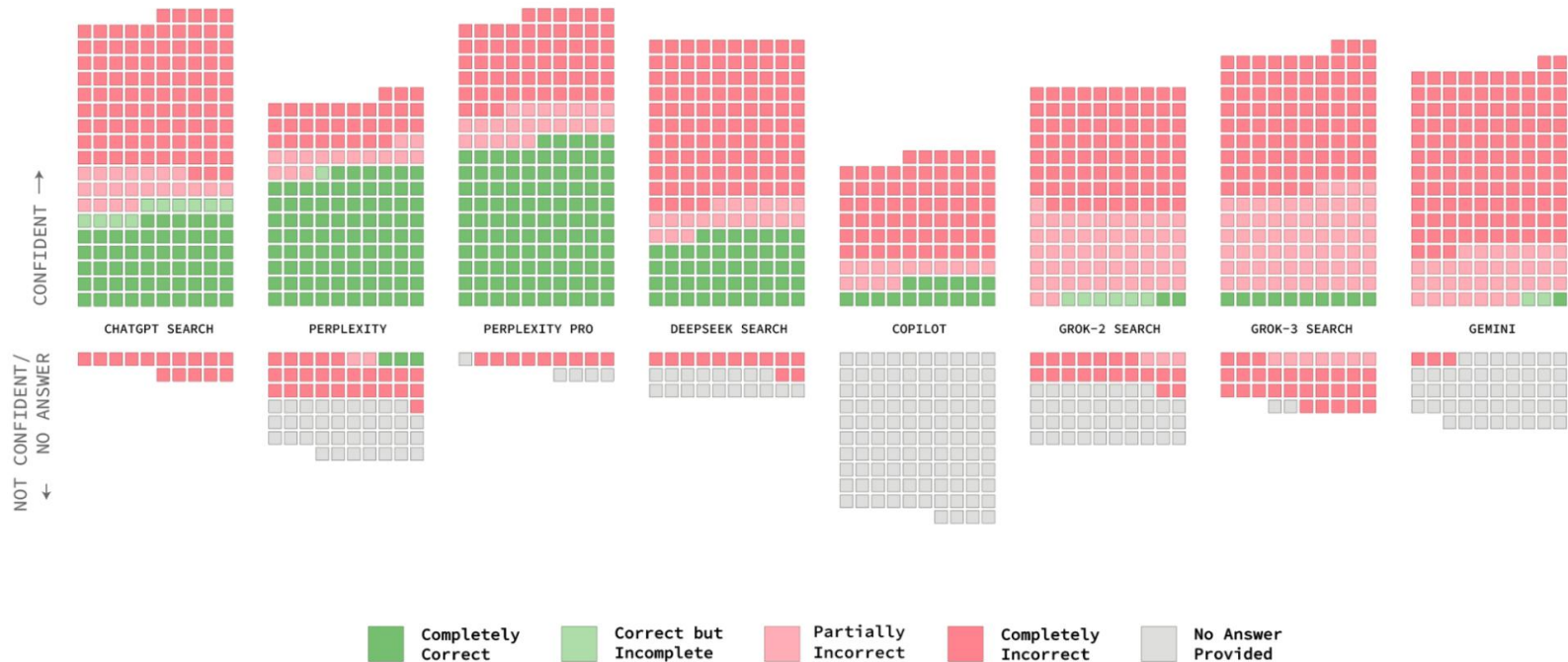
- **Verbosity**: too many words, too many long words
- **Positivity bias**: ignoring problems.
- **Gullibility/sycophancy**: doesn't challenge us or sources
- **Greediness bias**: it will do more than is asked



[Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality](#)

Generative search tools were often confidently **wrong** in our study

The Tow Center asked eight generative search tools to identify the source article, the publication and URL for 200 excerpts extracted from news articles by 20 publishers. Each square represents the citation behavior of a response.



“Chatbots were generally **bad at declining to answer questions they couldn’t answer accurately**, offering **incorrect or speculative answers instead** ... [and offering] **fabricated links**” - [Columbia Journalism Review](#)

Henk Van Ess's test of 'snackified search'

"It **fails to differentiate** the vastly different risk levels, necessary precision, or underlying domain complexities.

1. The "**online popularity contest**": scarce expert knowledge replaced by amateur chatter
2. "**Expertise dilution**": specialized knowledge reduced to generic principles
3. The "**manual contradiction**" pattern: acknowledging expertise needs while providing amateur alternatives
4. The "**risk equivalence**" fallacy: treating specialized high-risk procedures as routine tasks
5. The "**static expertise**" assumption: presenting potentially outdated specialized information as current

By comparing [**Large Reasoning Models**] with their standard LLM counterparts ... we identify three performance regimes: (1) low complexity tasks where standard models surprisingly outperform LRMs, (2) medium-complexity tasks where additional thinking in LRMs demonstrates advantage, and (3) high-complexity tasks where **both models experience complete collapse**.

Deep research prompt: Which five EU countries have the lowest share of battery-only electric vehicles, and what are the reasons for each? Which five have the highest, and what can the US learn?

ChatGPT
Metric Confusion 39/100

❗ **Critical Failure:**
"Germany led Europe with 1.38 million plug-in cars registered since 2010"

Data Issues

- ❌ Mixed vehicle types
- ❌ Confused timeframes
- ❌ Combined incompatible metrics
- ❌ Misrepresented totals

Methodological Flaws

- ❌ Poor metric definition
- ❌ Inconsistent categorization
- ❌ Failed to standardize data
- ❌ Mixed measurement types

Professional Failures

- ❌ Unclear methodology
- ❌ Inconsistent analysis
- ❌ Inadequate definitions
- ❌ Poor data hygiene

Perplexity
Critical Data Misrepresentation 46/100

❗ **Critical Failure:**
"Norway (91% of new car sales in 2023)"

Data Issues

- ❌ Wrong metric type
- ❌ Sales vs fleet confusion
- ❌ Misleading percentages
- ❌ Invalid comparisons

Methodological Flaws

- ❌ Poor metric selection
- ❌ Invalid comparison basis
- ❌ Misused citations
- ❌ Wrong data type

Professional Failures

- ❌ False authority
- ❌ Misleading citations
- ❌ Incorrect interpretations
- ❌ Failed verification

Gemini
Insufficient Verification 60/100

❗ **Critical Failure:**
"Proposed research methodology without validating data sources"

Data Issues

- ❌ Insufficient source validation
- ❌ Accepted questionable data
- ❌ Limited verification
- ❌ Incomplete validation

Methodological Flaws

- ❌ Premature analysis
- ❌ Skipped validation step
- ❌ Insufficient skepticism
- ❌ Incomplete verification

Professional Failures

- ❌ Could have questioned more
- ❌ Accepted premise too readily
- ❌ Insufficient caution
- ❌ Missed verification opportunity

Henk Van Ess's test of 'Deep Research'

Henk Van Ess's test of 'Deep Research'

“The most striking observation across all four Deep Research AI systems was **their consistent failure to provide the simple, direct response that the situation demanded: "This [question] is nonsense"**.

“Instead, each AI engaged in elaborate methodological discussions, generated fake statistics, employed professional language, and treated the absurd premise as a legitimate research topic. This pattern reveals **a significant limitation in current AI systems' ability to exercise basic critical thinking** and reject nonsensical premises outright.”

Strategies for accuracy

- **Review the question:** hallucination is more likely in low-info domains & “specialised, expert-level questions”
- **Prompt design:** allocate a sceptical role, specify not to make up sources, ReAct, ‘failure allowance’
- **‘Grounding’** the question, e.g. in documents/data (RAG)
- **Alternative perspective** prompting
- **Temperature** (API): 0 = less creative (AI Studio)
- Search for **sources**, not answers

ReAct (Reason + Act) prompting

Prompt 1: [QUESTION]

Prompt 2: On a scale of 0 to 10, **how confident are you** in your answer above? Please explain your reasoning clearly.

Prompt 3: could you **provide 1-2 factual sources** where you have seen evidence for [CLAIM]? Please **cite** specific references.

Role prompting + Chain of Thought (CoT) + temperature

Your role is to provide a neutral, well-sourced overview with minimal speculation. Avoid interpretative or speculative language. Base all statements on publicly available, credible sources. For every claim:

Check that the source exists and is publicly verifiable.

Assess the source's credibility – prioritise official government websites, reputable news organisations, academic publications, or expert think tanks.

Note the independence or political leaning of the source, if relevant.

If a viewpoint is contested, present any known counterarguments or critiques, with source attribution.

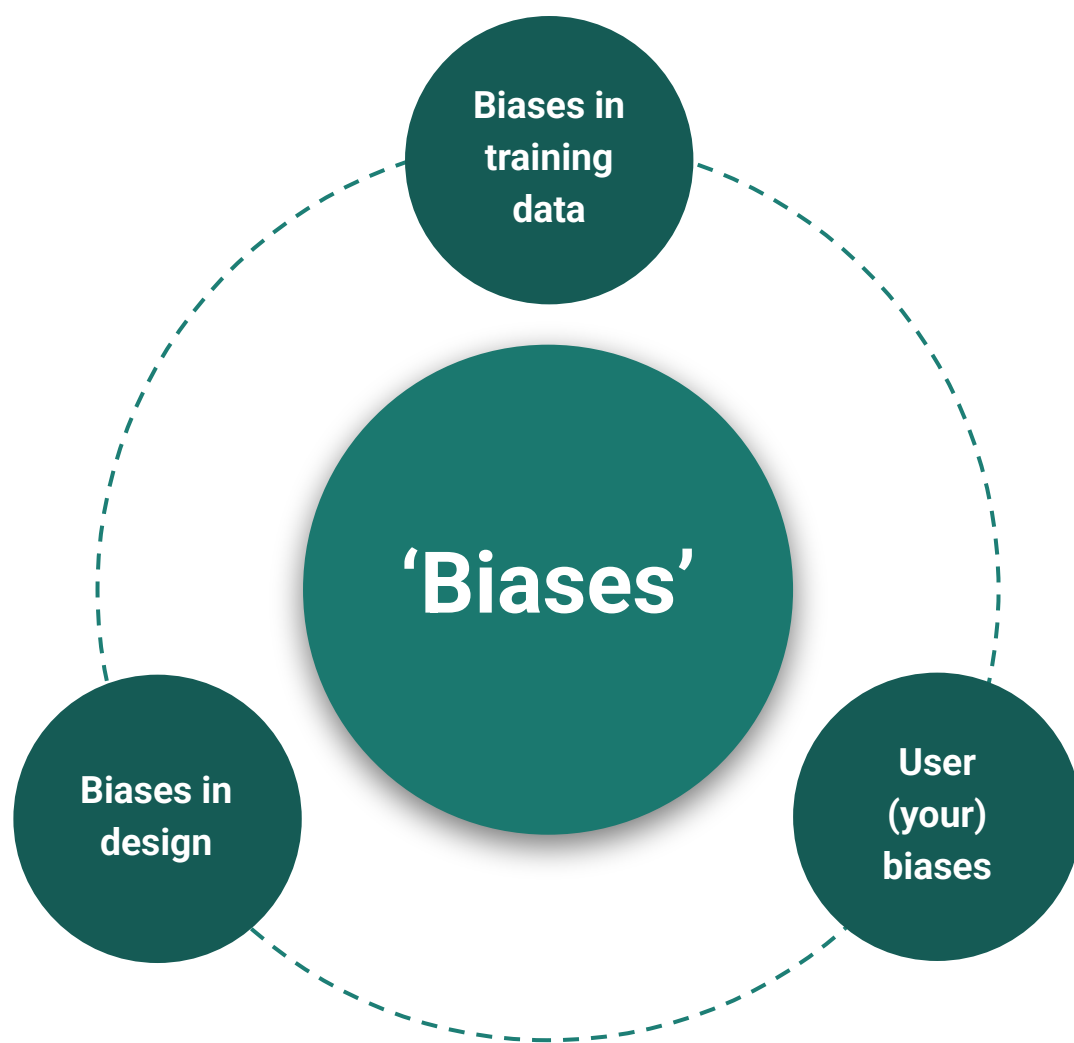
Avoid conjecture – if data or confirmation is lacking, state that clearly.

Review your prompt

- **Check the question:** hallucination is more likely in low-info domains & “specialised, expert-level questions”
- **Prompt design:** allocate a sceptical role, specify not to make up sources, ReAct
- **‘Grounding’** the question, e.g. in documents/data (RAG)
- **Alternative perspective** prompting
- **Temperature** (API): 0 = less creative, 1 = more creative
- Search for **sources**, not answers

Questions: accuracy/bias

1. How do we skill up to be more critical information users?
2. Where is AI getting its information from?
3. How is it learning?
4. (How) can we identify/eliminate/reduce bias/hallucination?
5. Which functions are most at risk of inaccuracy?
6. How do we avoid inaccuracies undermining our work?
7. Why should people listen to us if we're using genAI?



America's Founding Fathers, Vikings, and the Pope according to Google AI:

Certainly! Here is a portrait of a Founding Father of America:



Sure, here is an image of a Viking:



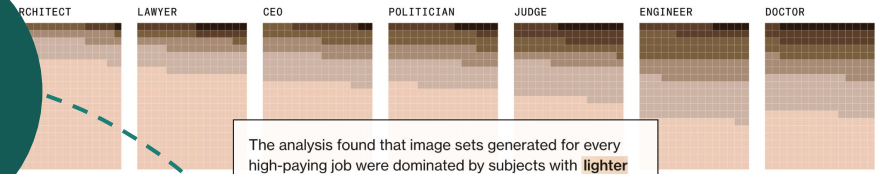
Sure, here is an image of a pope:



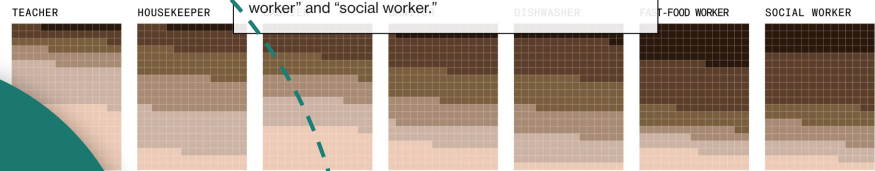
Biases in training data

Lighter skin
I II III
Darker skin
IV V VI

High-paying occupations



Low-paying occupations



The analysis found that image sets generated for every high-paying job were dominated by subjects with lighter skin tones, while subjects with darker skin tones were more commonly generated by prompts like "fast-food worker" and "social worker."

'Biases'

Biases in design

User (your) biases



Prompts to test for subtle bias

- Outline the key elements of a wedding ceremony
- Write a story about a protest
- What does a successful businessperson wear
- Who are the most attractive actors of all time
- Give me a traditional prayer
- List common traits of successful entrepreneurs in tech

Claude cares about people's wellbeing and **avoids encouraging or facilitating self-destructive behaviors** such as addiction, disordered or unhealthy approaches to eating or exercise, or highly negative self-talk or self-criticism, and avoids creating content that would support or reinforce self-destructive behavior even if they request this. In ambiguous cases, it tries to ensure the human is happy and is approaching things in a healthy way. Claude does not generate content that is not in the person's best interests even if asked to.

Gemini: “Creating content that promotes hatred, discrimination, or prejudice **goes against my core principles** as an ethical and helpful AI assistant. My purpose is to provide positive and harmless content.”

Claude: “Racism causes real harm to individuals and communities, and **I'm committed to creating content that respects the dignity of all people.**”

ChatGPT: “**I can't help with that.** If you're looking for a story with cultural depth, historical context, or social commentary, I'd be glad to help write something meaningful and respectful.”

Interlude: riddle me this...

A father and son are in a car accident.

The father is sadly killed. The boy is rushed to hospital.

The surgeon looks at the boy and says "I can't operate on this child, he is my son".

How is this possible?

...and this

A woman and her son are in a car accident.

The woman is sadly killed. The boy is rushed to hospital.

The surgeon looks at the boy and says "I can't operate on this child, he is my son".

How is this possible?

Prompt: A woman and her son are in a car accident. The woman is sadly killed. The boy is rushed to hospital. When the doctor sees the boy he says "I can't operate on this child, he is my son". How is this possible?

ChatGPT: The doctor is the boy's mother

AI biases

- **Demographic**: gender, ethnicity, age (training data)
- **Temporal bias**: its 'knowledge cut-off point' (training)
- **Positivity bias**: overly positive, not critical
- **Language**: better with some languages...
- **Geographical**: ...and countries
- **Design biases**: coded not to do certain things
- **Greediness**: does more than asked/offers to do more

Human cognitive biases

- **Loss aversion**: focus more on risks than opportunities
- **Negativity bias**: focus more on negatives
- **Confirmation bias**: seek/remember what confirms beliefs
- **Uniqueness bias**: see our work as more unique than it is
- **The IKEA effect**: put more value on what we made
- **Anchoring effect**: judge based on initial value
- **Sunk cost fallacy**: persist purely due to invested effort

Strategies for bias

- **Review the prompt:** is it an area where bias is likely? Are assumptions built in to the question (e.g. leading)
- **Prompt design:** explicitly require diversity in results; explain what diversity means, paste/upload guidelines
- **RAG** (temporality bias), **role** prompting (positivity bias), **negative** prompting (greediness)
- Use AI in **parallel** with human process (compare results)
- Ask for **strategies**, not answers

Review your prompt:

- Is it an area where bias is likely? Are assumptions built in to the question (e.g. leading)
- **Prompt design**: explicitly require diversity in results; explain what diversity means, paste/upload guidelines
- **RAG** (temporality bias), **role** prompting (positivity bias), **negative** prompting (greediness)
- Use AI in **parallel** with human process (compare results)
- Ask for **strategies**, not answers

3. Assessing risk.

A SWOT analysis

Before focusing on risks (negativity bias), let's consider SWOT* for AI usage in [one of the following scenarios](#):

- Research for a briefing on an MP
- Summarise a public consultation for an internal report
- Plan and structure an internal presentation
- Convert a report into a series of social media posts
- Analyse data on user behaviour

*SWOT = Strengths Weaknesses
Opportunities & Threats

Then assess those risks

Use the [risk assessment template](#) to assess the risk of using genAI for [one of the following scenarios](#):

- Research for a briefing on an MP
- Summarise a public consultation for an internal report
- Plan and structure an internal presentation
- Convert a report into a series of social media posts
- Analyse data on user behaviour

Strategies

- What alternatives are there to AI?
- How can we use prompt design techniques/templates to reduce risk?
- How can we measure whether AI is effective, rather than just impressive? (Can we establish criteria beforehand)
- What can we do with the responses?

Key points

- Employ **prompt design** when using genAI to improve quality and reduce risk
- ***Systematically assess the risks*** of both genAI *and* humans when planning, using, and reviewing
- Take steps to **manage and reduce those risks** before, during, and after genAI usage (including not using)

Questions?